

IMO Training – Number Theory

Level 2 Assignment 2

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Instructions:

- Write your solutions legibly on separate sheets of paper.
- Each question carries 7 marks.
- Complete solutions with justified arguments are required for full credit in every question.
- Both paper and electronic copies are accepted.
- Deadline for submission: 9 November, 2019

1. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all $m, n \in \mathbb{N}$

$$n + f(m) \mid f(n) + nf(m).$$

2. Find all positive integers x, y, z such that $x^5 + 4^y = 2013^z$.
3. Find all positive integers n such that $2^n + 12^n + 2011^n$ is a perfect square.
4. Are there integers a, b such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes?
5. For each positive integer $n \geq 2$, when 2^{2^n} is divided by $2^n - 1$, is the remainder always a power of 4?
6. Prove that there are infinitely many distinct pairs (a, b) of relatively prime integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

Classwork

IMO Shortlist 00N1,4; 02N1; 04N3; 05N6; 06N5; 07N1; 12N2; 13N1